

1. Cloud network Simulation and Visualization
 - a. Packet flow and routing visualization
 - b. Latency and throughput measurement
 - c. Congestion control simulation
 - d. Network topology design and visualization
2. Cloud-based Networking for Facebook class webapps/backends
 - a. Virtual network configuration (VPCs, subnets)
 - b. Load balancing and scalability
 - c. Secure Network architecture
 - d. Cloud DNS and CDN setup
 - e. Cloud-based monitoring (e.g., AWS CloudWatch, Azure Monitor)
3. Multi-Agent Communication Networks
 - a. Coordination and negotiation in agent networks
 - b. Fault-tolerant multi-agent communication
 - c. Routing of agent-generated data streams
4. Intelligent Network Management
 - a. Network log summarization
 - b. Automated alert classification
 - c. ChatOps for network management (e.g. Troubleshooting assistant)
5. Distributed Systems Networking
 - a. gRPC and REST-based microservice communication
 - b. Scalable secure API gateways
 - c. Data consistency across distributed nodes
 - d. Network partition handling in cloud networks (CAP theorem context)
 - e. Fault detection and recovery mechanisms in microservices
6. Edge and IoT Networking
 - a. Edge-to-cloud data routing
 - b. IoT device discovery and registration
 - c. Bandwidth optimization in constrained networks